CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 9

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Regression with Panel Data

Terminology: *Panel data* (or *longitudinal data*) consists of observations on the same n entities at two or more time periods T:

$$(X_{it}, Y_{it}),\ i = 1, 2, \ldots, n,\ \text{and}\ t = 1, 2, \ldots, T,$$

where
i refers to entity and t refers to time at it is observed.

If some observations are missing the panel is called *unbalanced panel*, otherwise - *balanced panel*.

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Regression with Panel Data: Entity Fixed Effects Regression

Let's assume that

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_i + u_{it},$$

where Z_i is an unobserved variable that depends on i but does not change over time t. Therefore, we can rewrite

$$Y_{it} = \alpha_i + \beta_1 X_{it} + u_{it},$$

where α_i are i-specific intercepts, referred as *fixed effects*.

Adding more variables of the two types, we get the Entity Fixed Effects Regression Model:

$$Y_{it} = \alpha_i + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \ldots + \beta_k X_{k,it} + u_{it}.$$

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Regression with Panel Data: Time Fixed Effects Regression

Similarly, we can assume that

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_t + u_{it},$$

where Z_t is an unobserved variable that depends on t but does not change with i. Therefore, we can rewrite

$$Y_{it} = \lambda_t + \beta_1 X_{it} + u_{it},$$

where λ_t are t-specific intercepts, referred as *time fixed effects*.

Adding more variables of the two types, we get the Time Fixed Effects Regression Model:

$$Y_{it} = \lambda_t + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \ldots + \beta_k X_{k,it} + u_{it}.$$

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Regression with Panel Data: Both Entity and Time Fixed Effects Regression

Similarly, we can assume that there are both Entity and Time Fixed Effects:

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \ldots + \beta_k X_{k,it} + u_{it}.$$

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Maximum Likelihood Estimation (MLE)

Maximum likelihood estimation:

Assume that a distribution can be parametrized by $\theta_1, \theta_2, \ldots, \theta_d$, then given a sample of i.i.d. $X_1, X_2, \ldots, X_n$ from this distribution $f(x|\theta_1, \theta_2, \ldots, \theta_d)$,
the unknown θ_k can be obtained by maximizing the likelihood function:

$$
\begin{aligned}
L(\theta_1, \theta_2, \ldots, \theta_d | X_1, X_2, \ldots, X_n) &= \underbrace{f(X_1, X_2, \ldots, X_n | \theta_1, \theta_2, \ldots, \theta_d)}_{\text{joint distribution}} \\
&= \prod_{k=1}^{n} f(X_k | \theta_1, \theta_2, \ldots, \theta_d).
\end{aligned}
$$

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MLE for Bernoulli Distribution

Example:
Consider a sample of i.i.d. $X_1, X_2, \ldots, X_n$ from Bernoulli(p), i.e. the pmf is

$$P(X = k|p) = \begin{cases} p^k(1-p)^{1-k}, & \text{if } k \in \{0, 1\}, \\ 0, & \text{otherwise.} \end{cases}$$

Then the Maximum Likelihood Estimators of p is

$$\hat{p}_{\text{MLE}} = \bar{X}.$$

MLE for Normal Distribution

Example:

Consider a sample of i.i.d. $X_1, X_2, \dots, X_n$ from Normal(μ, σ^2), i.e. the pdf is

$$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R},$$

Then the Maximum Likelihood Estimators of μ and σ^2 are

$$\begin{aligned} \hat{\mu}_{\text{MLE}} =& \bar{X}, \\ \hat{\sigma}^2_{\text{MLE}} =& \frac{1}{n}\sum_{k=1}^{n}(X_k - \bar{X})^2. \end{aligned}$$

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Consistency of MLE

Thm.
Under appropriate smoothness conditions on f, the MLE $\hat{\theta}_n$ from an i.i.d. sample of size n is consistent, that is, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0,$$

where θ denotes the true value of the parameter.

Asymptotic Distribution of MLE

Thm.
Under smoothness conditions on f, as $n \to \infty$,

$$\sqrt{nI(\theta)}(\hat{\theta} - \theta) \xrightarrow{D} N(0,1),$$

where θ denotes the true value of the parameter and

$$I(\theta) = -\mathrm{E}\left[\frac{\partial^2}{\partial\theta^2} \ln f(X|\theta)\right].$$

$I(\theta)$ is called *Fisher information*.

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Regression with Binary Dependent Variable: Linear Probability Model

Linear Probability Model

Let Y_i be a binary variable. The linear probability model is the following multivriate regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \ldots + \beta_k X_{ik} + u_i \quad \text{with} \quad i \in \{1, 2, \ldots, n\}.$$

Regression with Binary Dependent Variable: Linear Probability Model

Linear Probability Model

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Note that

- $\mathrm{E}[Y|X_1, X_2, \ldots, X_k] = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \ldots + \beta_k X_{ik}$, if exogenous.
- $\mathrm{E}[Y|X_1, X_2, \ldots, X_k] = P(Y = 1|X_1, X_2, \ldots, X_k)$.
- $\boldsymbol{\beta} = [\beta_0, \beta_1, \ldots, \beta_k]^\mathsf{T}$ can be estimated via OLS.
- Errors are always heteroskedastic.
- The slope coefficient β_j is the change in probability of $Y = 1$, given $\Delta X_j = 1$.
- Is predicted probability between 0 and 1? Exogeneity?

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Generalized Linear Model (GLM)

The Linear Probability Model,

$$\mathrm{E}[Y|X_1, X_2, \ldots, X_k] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k,$$

can be generalized via introducing a so called *link function* $g(p)$, which is assumed to be invertible, as follows:

$$\mathrm{E}[Y|X_1, X_2, \ldots, X_k] = g^{-1}\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k\right).$$

Generalized Linear Model (GLM)

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Equivalently,

$$g\left(\mathrm{E}[Y|X_1, X_2, \ldots, X_k]\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k.$$

Generalized Linear Model (GLM)

Given Generalized Linear Model

$$\mathrm{E}[Y|X_1, X_2, \ldots, X_k] = g^{-1}\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k\right),$$

- $\boldsymbol{\beta} = [\beta_0, \beta_1, \ldots, \beta_k]^\mathsf{T}$ are estimated via Method of Maximum Likelihood.
- The Maximum Likelihood Estimators (MLE) can be shown to be consistent.
- MLE can be shown to follow normal distribution, if n is large.
- Thus, under assumption of large n, z and F tests can be used.
- If $\Delta X_j = 1$ for some j, while holding other variables fixed, we get

$$\begin{aligned}\Delta\hat{Y} &= \mathrm{E}[Y|\ldots, X_j + 1, \ldots] - \mathrm{E}[Y|\ldots, X_j, \ldots] \\ &= g^{-1}\left(\beta_0 + \ldots + \beta_j(X_j + 1) + \ldots \beta_k X_k\right) - g^{-1}\left(\beta_0 + \ldots + \beta_j X_j + \ldots \beta_k X_k\right),\end{aligned}$$

which cannot be simplified further.

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Probit and Logit Regression

The GLM

$$\mathrm{E}[Y|X_1, X_2, \ldots, X_k] = g^{-1}\left(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k\right),$$

is called

- Probit if

  $$g^{-1}(z) = \Phi(z) \doteq \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

  is the cdf of a standard normal random variable. g in this case is called *probit* function and denoted by $\mathrm{probit}(p)$.

- Logit (or Logistic regression) if

  $$g^{-1}(z) \doteq \frac{1}{1+e^{-z}}.$$

  In this case, g is called *logit* function and denoted by $\mathrm{logit}(p) = \ln\left(\frac{p}{1-p}\right)$.

Probit and Logit Regression

Example:

```
n = 200
beta0 = -5
beta1 = 10

X = runif(n)
p = inv.logit(beta0 + beta1*X)
Y <- vector()
for (i in 1:n){Y[i] = rbinom(n = 1, size =1, p[i])}
df = data.frame(X,Y)
```

Logit:

```
fit <- glm(Y ~ X, data = df, family = "binomial")
summary(fit)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -4.7839     0.6559  -7.293 3.02e-13 ***
X             9.1660     1.1842   7.740 9.92e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 277.18  on 199  degrees of freedom
Residual deviance: 141.43  on 198  degrees of freedom
AIC: 145.43

Number of Fisher Scoring iterations: 5
```

Probit:

```
fit <- glm(Y ~ X, data = df, family = binomial(link = "probit"))
summary(fit)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.8274     0.3494  -8.092 5.89e-16 ***
X             5.4109     0.6214   8.707  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 277.18  on 199  degrees of freedom
Residual deviance: 139.56  on 198  degrees of freedom
AIC: 143.56

Number of Fisher Scoring iterations: 6
```